

Diksha Damahe

B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)

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LinkedIn | GitHub | ORCID ID | Scopus ID

Profile

B.Tech CSE (AI & ML) student at VIT Bhopal with hands-on experience in **machine learning**, **data-driven applications**, and **full-stack development**. Skilled in **Python**, **Java**, **JavaScript**, **TypeScript**, and frameworks like **React**, **Next.js**, **Django**. Built production-style applications including a secure password manager, landing platforms for ed-tech and defense domains, an ANN-based earthquake classifier, and a campus food-ordering system. Strengthened cloud foundations through AWS certifications and improved backend proficiency with advanced SQL training.

Education

VIT Bhopal University, Bhopal, Madhya Pradesh

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning), CGPA: **8.06**
2023 – 2027

Experience

Hindustan Copper Limited (HCL), Malanjkhand, Madhya Pradesh

June 2026 – Present

Software Engineering Intern (AI/ML)

- Developed and deployed **RecovAI**, an AI-based predictive analytics and industrial process monitoring system for the flotation concentrator plant, enabling real-time shift recovery forecasting.
- Designed anomaly detection pipelines and recommended optimal chemical dosages based on ore feed parameters, providing operators with data-driven decision support to prevent mineral recovery losses.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C++, SQL

Frameworks & Libraries: React.js, Next.js, Django, Flask, Node.js, TensorFlow, Keras, scikit-learn, Pandas, NumPy

Cloud & DevOps: AWS (EC2, S3, EBS), Cloud Computing Fundamentals, Git, GitHub, Postman

Databases: PostgreSQL, MongoDB, Advanced SQL Queries

AI / ML: ANN, Supervised Learning, Feature Engineering, Data Processing, Model Evaluation (Accuracy, Precision, Recall, F1-Score)

Web & UI/UX: Tailwind CSS, Framer Motion, Responsive Design, Component Architecture

Core CS: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS

Languages: Hindi (Fluent), English (Fluent)

Projects

RecovAI — AI-Powered Copper Recovery Monitoring

GitHub | Live Demo

Python, Streamlit, XGBoost, Isolation Forest, scikit-learn, Pandas, Matplotlib, SHAP

- Developed an industrial dashboard analyzing 18+ months of metallurgical data to forecast recovery rates with an **R² of 0.9720** and **MAE of 0.1042%**.
- Engineered a Reagent Dose Intelligence Engine to optimize SIPX, Frother, and pH levels, and integrated an Isolation Forest model to detect process anomalies.

Chemical Equipment Parameter Visualizer (Web + Desktop)

GitHub | Live Demo

Django, Django REST Framework, React, PyQt5, Chart.js, Matplotlib, Pandas, ReportLab

- Built a hybrid analytics system where users upload CSV equipment datasets and view synchronized insights on both web and desktop clients.

- Implemented chart-rich dashboards (bar, line, doughnut) and summary cards to improve readability, visual hierarchy, and decision support.
- Developed authenticated dataset history, API-backed analytics endpoints, and downloadable PDF reports for structured data communication.
- Deployed full stack to production (Vercel + PythonAnywhere) with a Windows desktop executable for cross-platform accessibility.

NAFLD Risk Stratification Pipeline (Research Project)

GitHub

Python, scikit-learn, XGBoost/LightGBM/CatBoost, SHAP, Matplotlib, NHANES

- Designed and benchmarked 24 classification algorithms on NHANES-derived features for population-level NAFLD risk screening.
- Built a reproducible preprocessing and validation workflow with stratified splitting, leakage-safe SMOTE, and cross-validation.
- Produced comparative ROC and feature-importance visualizations plus SHAP explainability outputs for interpretable visual reporting.
- Achieved best model performance of 0.9679 ROC-AUC (AdaBoost) on proxy risk labels.

Aegis Vault – Secure Product Dashboard Design

GitHub | Live Demo

Next.js, TypeScript, MongoDB, Prisma, NextAuth, Tailwind CSS, Framer Motion

- Built a secure, highly interactive dashboard experience with search, category filters, and structured card-based data presentation.
- Integrated client-side encryption workflows with clear visual states, improving usability for sensitive information handling.
- Designed polished UI behavior (modals, transitions, theme controls) to communicate complex actions with minimal cognitive load.

Veil — Minimalist Android Streaming Aggregator

GitHub

Flutter, Dart, ExoPlayer, TMDB API, TypeScript

- Developed a cross-platform Android streaming media aggregator leveraging **TMDB API** for metadata and **ExoPlayer** for video playback.
- Integrated a self-hosted custom resolver (cinempro-org/core OMSS) to dynamically aggregate and resolve stream sources.

Additional Projects (Selected):

- **Shruti – Neumorphic Bilingual Audio-Text Syncer** (Flutter, Dart, Riverpod, Firebase, Firestore, Whisper AI): bilingual mobile app using Whisper AI to synchronize Osho discourses with real-time text. [GitHub](#)
- **IELTS Precision & Progress** (React, TypeScript, Vite, Tailwind CSS, Framer Motion): conversion-focused platform with metrics, testimonials, and micro-interactions. [GitHub](#) | [Live Demo](#)
- **VyomGarud – Counter-UAV Defense Website** (Next.js, React, Tailwind, Framer Motion): cinematic, data-led interface design with animated metrics and responsive visual storytelling. [GitHub](#) | [Live](#)
- **Hostel-Bites – Meal Management Platform** (JavaScript, Firebase, Firestore): real-time order tracking and operational dashboard workflows for hostel users and admins. [GitHub](#) | [Live](#)

Publications

Secure Phishing Analysis and Robust Training Algorithm (SPARTA) for Phishing Website Detection

Diksha Damahe, Ajay Kumar Phulre

2026 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS), Bhopal, India

Detection rate: **99.15%** on PhishTank and UCI benchmarks; ensemble-based phishing detection framework.

Certifications

- Introduction to Internet of Things — Elite NPTEL Online Certification, IIT Kharagpur (Jan–Apr 2026)
- Cloud Computing — Elite NPTEL Online Certification, IIT Kharagpur (Jan–Apr 2025)
- Google IT Support Certificate - Google (Jan 2026)
- Applied Machine Learning in Python — University of Michigan (Nov 2025)
- AWS Certification Course — Intellipaat (Sep 2025)
- AWS Technical Essentials — Amazon Web Services (Sep 2025)
- Mastery in JavaScript — Lernx (Aug 2025)
- Advanced SQL for Technical Interviews — Bosscoder Academy (Feb 2025)
- Creating Apps in Python — Lernx (May 2024)
- Python 3 Bootcamp — Lernx (May 2024)
- Fundamentals of AI and ML — Vityarthi (Jan 2024)
- Python Essentials — Vityarthi (Jan 2023)

Extracurricular & Leadership

Content Team Leader, FinTech Club — VIT Bhopal University

- Leading the content team, managing content strategy, communication, and execution for FinTech-related events and initiatives.

Volunteer, National Service Scheme (NSS) — VIT Bhopal University

- Supported university events and community engagement activities through NSS initiatives.